Market Expectations from Options: Extracting and Interpreting Risk-Neutral Distributions

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Abstract

This paper forms the foundation of a personal project investigating the extraction and interpretation of risk-neutral distributions (RNDs) from options market data, motivated by their role within an Entropic Martingale Optimal Transport (EMOT) framework. Critically, only basic undergraduate-level mathematics and economics were used for any derivations. Moving beyond implementation, the paper treats RNDs as economically meaningful objects that encode both market expectations and investor risk preferences. Using QQQ options data, RNDs are recovered via the Breeden–Litzenberger methodology through a carefully constructed computational pipeline involving data filtering, smoothing, interpolation, and numerical differentiation under strict no-arbitrage conditions.

Empirically, the extracted distributions exhibit a strong and consistent pattern of negative skewness, observed in 78.2% of valid cases across maturities ranging from 10 to 64 days. This finding is robust across the subset of high-quality option data that satisfies both statistical and economic validation criteria.

To interpret this asymmetry, the paper draws on the Consumption-Based Capital Asset Pricing Model (CCAPM), showing how risk-neutral probabilities overweight adverse states due to the higher marginal utility of consumption during downturns. This framework provides a coherent explanation for both the observed skewness in RNDs and the elevated prices of downside protection instruments such as long put options, which serve as insurance against low-consumption states.

Overall, the project demonstrates how market-implied distributions, when carefully extracted, offer insight into the economic forces shaping asset prices, and critically, without having to resort to advanced mathematics and economics for a logical explanation.

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1 Introduction and Motivation

1.1 Project Context and Motivation

This project emerged from implementing an Entropic Martingale Optimal Transport (EMOT) pipeline alongside the NUS Investment Society's Quantitative Finance (NUSiQF) options team. EMOT is an advanced algorithm for portfolio optimization that requires a risk-neutral probability distribution as a critical input. To support this pipeline, we implemented a system to extract risk-neutral distributions from real options market data using the Breeden-Litzenberger methodology as seen in (Breeden and Litzenberger, 1978).

After completing the initial EMOT implementation, I wished to deepen my understanding of the risk-neutral distribution component—both its theoretical foundations and its empirical properties. This project therefore shifts its focus from the original EMOT pipeline: rather than merely serving as a component of EMOT, I proceed to treat RND extraction as an independent investigation into what market option prices reveal about investor beliefs and risk preferences.

1.2 Pedagogical Approach

This paper is written, at the time of writing, by a junior undergraduate student with a basic background in calculus, probability, options pricing models and stochastic calculus. I have therefore made the conscious effort to rely on logical intuition and basic mathematics as much as possible throughout this paper, such that a fellow junior undergraduate student with only basic calculus and probability knowledge would be able to follow along without much (if any) issue. As a result, rigour may at times be sacrificed for clarity.

1.3 AI Assistance

In supporting this paper, Claude AI (Anthropic) and ChatGPT (OpenAI) were utilized in the following capacities:

- **Code generation**: Implementation of helper functions, grid construction, and visualization routines
- **Mathematical exposition**: Clarification and explanation of CCAPM concepts and derivations for undergraduate accessibility
- **Report structure**: Organization and phrasing of sections
- **Data analysis**: Assistance in interpreting statistical patterns and connecting empirical findings to theory

All computational work (running the pipeline, extracting RNDs, validating results) and economic interpretation of findings are my own. AI was not used for the core derivations or theoretical insights which represent my own independent learning.

1.4 Terminology

Throughout this paper, we use **RND** (risk-neutral distribution) to denote the probability distribution of terminal prices under the risk-neutral measure $\mathbb{Q}$, reserving "risk-neutral density" for the mathematical function itself. This distinction, while subtle, clarifies discussions where both the distribution (as a concept) and its density function (as a mathematical object) appear.

1.5 Document Structure

The paper is organised as follows:

1. **Asset Pricing Foundations**: We introduce risk-neutral distributions informally, then build the theoretical foundation through CCAPM. We derive the risk-neutral measure $\mathbb{Q}$ from first principles, explaining why option prices contain information about probability distributions beyond the physical world.
2. **Data and Methodology Outline**: We describe the QQQ options dataset and provide an overview of the Breeden-Litzenberger procedure.
3. **Technical Implementation**: We detail each step of the computational pipeline: constructing strike-call grids, smoothing functions, applying numerical differentiation, and post-processing distributions.
4. **Results and Analysis**: We present aggregate statistics and showcase representative distributions, highlighting the pronounced negative skewness observed across 78.2% of successful RNDs.
5. **Economic Interpretation**: We return to CCAPM to explain why market-extracted distributions are negatively skewed, connecting this to rational hedging demand for put options during adverse states.
6. **Limitations**: We discuss data constraints, filtering stringency, and alternative modeling approaches to contextualise the scope and reliability of our findings.
7. **Conclusion**: We provide a summary of our findings throughout this paper, a brief return to the original EMOT pipeline, and finally closing thoughts on the project as a whole.

2 Asset Pricing Foundations

2.1 Introduction to RNDs

A RND is the probability distribution of an underlying asset's future, or terminal, price, under the risk-neutral measure $\mathbb{Q}$. Typically, options of the underlying in question are used to arrive at this distribution through the Breeden-Litzenberger methodology as described in Breeden and Litzenberger (1978).

Unlike a physical probability distribution, the RND incorporates investors' attitudes toward risk and therefore reflects both beliefs and preference hence the need for a risk-neutral measure $\mathbb{Q}$. If so, then how does $\mathbb{Q}$ differ from the physical measure $\mathbb{P}$ and why does an RND differ from the true probabilities of future states?

2.2 Consumer-based Capital Asset Pricing Model (CCAPM)

To understand this transformation from $\mathbb{P}$ to $\mathbb{Q}$, it is useful to begin with CCAPM which links asset prices to investors' marginal utilities of consumption. Throughout this introduction to CCAPM, I will routinely refer to Cochrane (2005).

2.2.1 Utility Function

Our first objective would be to model a stream of uncertain cash flows. Let us first find the value at time t of a *payoff* X_{t+1} which is the payoff at the next period, which includes both the stock price and dividend. The *payoff* is not to be confused with the *profit* or *return*; X_{t+1} denotes the investment value at that time without subtracting or dividing by the cost of investment.

To find the payoff value, we require a mathematical formalism describing an investor's wants which is how we arrive at the *utility function* defined over current and future values of consumption:

$$U(c_t, c_{t+1}) = u(c_t) + \beta \ \mathbb{E}_t[u(c_{t+1})]$$

where c_t denotes consumption at time t.

The utility function is an increasing one, capturing the fundamental desire for more consumption. Further, it is concave, reflecting the declining marginal value of additional consumption. Additionally, the subjective discount factor β represents time preference — how much less a person values future consumption compared to current consumption.

2.2.2 Basic Pricing Equation

Denoting by e the original consumption level, and by ξ the amount of asset chosen to buy, then an investor would want to maximise his utility for his current consumption, and the expected utility for his future consumption:

$$\max\{u(c_t) + \mathbb{E}_t[\beta \ u(c_{t+1})]\} \text{ s.t.}$$

$$c_t = e_t - P_t\xi$$
$$c_{t+1} = e_{t+1} + X_{t+1}\xi$$

Substituting the constraints and setting the derivative with respect to ξ to zero, we obtain

$$P_t u'(c_t) = \mathbb{E}_t[\beta u'(c_{t+1})X_{t+1}] \tag{1}$$

which can be manipulated into

$$P_t = \mathbb{E}_t\left[\beta \frac{u'(c_{t+1})}{u'(c_t)} X_{t+1}\right] \tag{2}$$

From (1), the $P_t u'(c_t)$ term is the loss in utilty if the investor buys another unit of the asset; correspondingly, $\mathbb{E}_t[\beta u'(c_{t+1})X_{t+1}]$ is the expected increase in utility he obtains from the extra payoff at $t+1$. In other words, if the investor consumes more today, his utility today correspondingly increases at the cost of a reduced utility tomorrow.

2.2.3 Stochastic Discount Factor

A convenient way to decompose (2) is to define the stochastic discount factor M_{t+1}:

$$M_{t+1} = \beta \frac{u'(c_{t+1})}{u'(c_t)} \tag{3}$$

which leads to

$$P_t = \mathbb{E}_t^{\mathbb{P}}[M_{t+1}X_{t+1}] \tag{4}$$

which is purposely denoted with $\mathbb{P}$ to denote that we are currently under the physical world probability measure, and to contrast against similar equations introduced later that are under $\mathbb{Q}$.

The stochastic discount factoris not merely a convenient term introduced here — it is a concept used throughout major asset pricing models and not just in CCAPM, as shown in Smith and Wickens (2002).

2.3 State Prices

While (4) provides a compact expression for asset prices, it obscures how individual future states contribute to valuation. Since risk-neutral probabilities ultimately arise from the pricing of contingent claims across states of the world, it is useful to rewrite the pricing equation in a state-dependent form which will eventually lead to the derivation of the $\mathbb{Q}$ probability measure.

2.3.1 Conditional Expectation

For simplicity, we will work in a discrete probability space with n possible outcomes $\omega_1, \omega_2, \cdots, \omega_n$. For any random variable Y, we would have

$$\mathbb{E}[Y] = \sum_i Y(\omega_i) \Pr(\omega_i)$$

or, conditional on time t,

$$\mathbb{E}_t[Y] = \sum_i Y(\omega_i) \Pr(\omega_i | \mathcal{F}_t), \tag{5}$$

where $\mathcal{F}_t$ denotes a filtration and represents all information known up to time t.

Applying (5) to (4), we have

$$P_t = \sum_i M_{t+1}(\omega_i) X_{t+1}(\omega_i) \Pr(\omega_i | \mathcal{F}_t) \tag{6}$$

2.3.2 State-dependent Notations

(6) is clearly cumbersome. We therefore assign the following notations:

- $p_i = \Pr(\omega_i | \mathcal{F}_i)$, denoting the probability assigned to the ith state
- $m_i = M_{t+1}(\omega_i)$, denoting the SDF value assigned to the ith state
- $x_i = X_{t+1}(\omega_i)$, denoting the payoff assigned to the ith state

Substituting these notations into (6), we obtain

$$P_t = \sum_i p_i m_i x_i \tag{7}$$

2.4 Derivation of Q-measure

With (7), we now actually have what we need to finally derive, and rationalise the existence of, the $\mathbb{Q}$ probability measure.

2.4.1 q-weights

From (7), we can see that each state's probability is weighted by the stochastic discount factor-which in turn reflects the value investors place on consumption in that state. We next aim to make this weighting explicit, which also has the side effect of creating an even more convenient notation.

Referencing the methodology provided in the StackExchange post by LocalVolatility (2017), we first define the weight

$$q_i = \frac{p_i m_i}{\sum_j p_j m_j}$$

where

$$\sum_i q_i = 1, \tag{8}$$

informing us that $\mathbb{Q}$ is in fact a probability measure.

2.4.2 Risk-free rate

Next, we introduce the risk-free rate r. Suppose \$1 is invested today at time t in a risk-free asset, yielding $1+r$ at time $t+1$ regardless of which ith state ω_i occurs. We would naturally have $x_i = 1+r$ for any ith state. Therefore for an asset with current price \$1,

$$1 = \sum_i p_i m_i (1+r),$$

which is manipulated into

$$\sum_i p_i m_i = \frac{1}{1+r}. \tag{9}$$

2.4.3 Derivation

Substituting (9) and (8) into (7), we have

$$P_t = \sum_i p_i m_i x_i = \frac{1}{1+r} \sum_i q_i x_i$$

which by definition of expectation can be further simplified into

$$\frac{1}{1+r} \mathbb{E}_t^{\mathbb{Q}}[X_{t+1}]$$

where $\mathbb{Q}$ is introduced to clearly denote the new probabilty measure, and t notations are reintroduced to mark the shift away from the previous state-dependent ones.

2.5 Relation to Options

Having established that asset prices can be expressed as discounted expectations under the risk-neutral measure, the pricing problem is transformed from one involving investors' preferences and marginal utility into one involving a probability measure that is implicitly embedded in market prices.

However, the risk-neutral measure itself is not directly observable. Instead, it must be inferred from the prices of traded securities whose values reflect market expectations of future states. Option markets are particularly well suited for this purpose, as option prices contain information about the valuation of payoffs across a wide range of future asset prices. Recovering the risk-neutral distribution therefore requires an empirical procedure that works backwards from observed option prices to infer the probability distribution consistent with those prices.

3 Data and Methodology Outline

Before outlining this extraction methodology, the following section describes the option market data and supporting inputs used in the analysis.

3.1 Raw Data

3.1.1 Sources

Data	**Source**
Options	EODHD, 2-year daily anchors
Spots	YahooFinance
Federal Funds Effective Rates	Federal Reserve Bank
Quarterly Dividends	MacroTrends

3.1.2 Rationale for N=90

We have tested only the first $N = 90$ snapshot dates from the two years worth of original options data as the goal had shifted from providing a prior to EMOT to generating presentable empirical distributions — as such, priorities were changed. The following highlights the main factors taken into consideration:

1. Anything more than $N = 90$ would be too computationally expensive
2. The full two years worth of data cannot be uploaded onto Github without purchasing additional storage
3. $N = 90$ covers approximately three months, which is an entire quarter. This in turn will likely cover multiple market regimes
4. It remains a trivial task to extend the same methodology to the remainder of the dataset

3.2 Metrics

Metric	**Value**
Total unique strikes	768
Strike range	\$129.78–\$830
Total snapshot-expiry combos	4009
Median bid-ask spread	$\sim$\$0.015
Valid bid-ask pairs	$\sim$95%

3.3 Methodology Outline

We next provide a rough outline for the pipeline, implemented entirely in Python, before going in-depth into each step in the next section.

1. Build a strike-call grid, prioritisng OTM options and using put-call parity to recover where necessary to recover call prices. When appending values to the grid, monotonocity and convexity checks are done to ensure that no-arbitrage constraints are not violated before a Savitzky-Golay filter is applied to remove outliers
2. Next, a call price function is created from the grid, smoothened with a Savitzky-Golay filter again utilising an adaptive window sizing strategy.Then, the function was interpolated with a Piecewise Cubic Hermite Interpolating Polynomial
3. Finally, the Breeden-Litzenberger formula was used to recover distributions from the second derivative of the function. A Kernel Distribution Estimation utilsing Gaussian curves was applied to smoothen the final distribution before being normalised to a total area of 1

4 Pipeline Implementation

4.1 Strike-Call grid

We first build a grid consisting of strike price - call price pairs for a given snapshot date and expiration date pair, by appending valid pairs to a `list`.

Out of the Money (OTM) options are prioritised over In the Money (ITM) ones because of the former's higher liquidity. This in turn implies that more information can be extracted from OTM options because of larger volumes and tighter bid-ask spreads.

However, not all snapshot dates present call prices which leads us to the put-call parity.

4.1.1 Put-Call Parity

The put-call parity defines the relationship between the prices of a European call and put option that share the same underlying, strike, and expiration dates, and is given as:

$$C = P + e^{-rT}(F - K) \quad \text{where} \quad F = Se^{(r-q)t},$$

and q is the dividend, calculated on a quarterly basis.

This provides us with a method to recover call prices from put data, in the absence of the former, through implied futures prices. This is the same strategy documented in Ait-Sahalia and Lo (1998) and subsequently seen in Smolski (2024).

Once the grid is formed, we proceed to remove obvious outliers.

4.1.2 Outlier Detection

The Savitzky-Golay filter is first applied to the formed grid of call price - strike pairs to establish a baseline and subsequently in identifying possible outliers. This technique was chosen due to its ability to maintain the integrity of the original signal during smoothing, which is accomplished through the use of linear least squares, as explained in Savitzky and Golay (1964). Additionally, a low-degree polynomial is chosen for simplicity and to prevent overfitting.

After a baseline is established, residuals are calculated to identify outliers. Points with residuals exceeding 1.5σ are flagged as outliers and removed from the grid. This threshold retains approximately 87% of data under a normal distribution, balancing outlier removal against the need to preserve data points that reflect true market conditions.

4.2 Forming the call price function

We next proceed with forming a smoothened function $C(K)$ for use with the Breeden-Litzenberger formula later.

4.2.1 Smoothening with Savitzky-Golay

Savitsky-Golay is again applied to the grid to further smoothen it before interpolation. Unlike the first iteration when removing outliers, an adaptive window sizing strategy is now utilised which assigns windows based on grid size:

- 5–7 points: Window = 5
- 8–19 points: No. of points $*$ 0.45
- $\geq$ 20 points: No. of points $*$ 0.40

Additionally, the window size is required to be an odd integer so that the local polynomial is centered, with an equal number of points on either side. Therefore, a low polynomial degree of 2–3 is chosen.

4.2.2 Interpolation with PCHIP

Interpolation of the curve is then achieved through a Piecewise Cubic Hermite Interpolating Polynomial (PCHIP), which creates a smooth, continuous function $C(K)$ by strictly connecting every data point.

PCHIP was chosen as an interpolator due to its ability to maintain monotonicity of the function - this is a no-arbitrage condition that must be satisfied (explained below).

4.2.3 No-arbitrage Constraints

Valid RNDs must satisfy no-arbitrage constraints, leading directly to the convexity condition of $C(K)$:

$$C(K) \geq 0 \qquad \frac{\partial C}{\partial K} \leq 0 \qquad \frac{\partial^2 C}{\partial K^2} \geq 0$$

Therefore, the second derivative must be decreasing (non-strictly). Otherwise, butterfly spreads can be utilised to abuse arbitrage.

The `mono_convex()` validating function during grid construction prevents violations.

4.3 Breeden-Litzenberger formula

The formula, as described in Breeden and Litzenberger (1978), is given by

$$f(K) = e^{rT} \frac{\partial^2 C}{\partial K^2}$$

where $C(K)$ is the smoothened function from before.

An array of evenly-spaced strikes is first formed based on the smallest and largest strike values from the original grid before $C(K)$ is differentiated against that array to the second order. The negative values are floored to 0, before the new function is multiplied by the discount factor e^{rT} and subsequently normalised to maintain an integral of 1 to maintain the property of probability distribution functions (PDFs).

4.3.1 Kernel Distribution Estimation

Following from the Breeden-Litzenberger formula, we realise that the resulting distrbution consists of many irregular peaks in the observed RNDs. We counteract this through Kernel Distribution Estimation (KDE) smoothing with Gaussian curves (or kernels) as described in Silverman (1986), which we are able to conveniently utilise through the `Scipy` module.

First, we normalise the RND distribution to obtain KDE weights based on the entire sum of the PDF. Together with the array of evenly-spaced strikes from before, `gaussian_kde` is applied while using Scott's method by default in estimating the optimal kernel width for use.

As with before, any negative values are floored to zero, and the result is again normalised to maintain the PDF property of an integral of 1.

If any distribution cannot be normalised at any step, it is rejected.

5 Results & Analysis

5.1 Dataset Processing Summary

Metric	**Value**
Total snapshot-expiry combinations	7353
Successfully processed	606 (8.2%)
Rejected	6747 (91.8%)
Median grid size (final)	40 strikes
Min/Max grid size	8–97 strikes
Mean grid size	38.4 strikes
Median DTE	17 days
DTE range	10–64 days

5.1.1 RND Shape Metrics

Metric	Value
Median skewness	-0.8974
% negatively skewed	78.2%
Median excess kurtosis	1.5981
Median volatility (std dev)	12.45

Key Observation: Across 7353 snapshot-expiration combinations, 606 successfully processed RNDs represent the subset satisfying simultaneous requirements: (1) sufficient strike coverage with valid bid-ask pairs, (2) passing no-arbitrage tests, (3) retaining $\geq$8 grid points post-filtering, and (4) successful numerical normalisation. Critically, the 606 successful RNDs span the full 10–64 day maturity range, demonstrating the pipeline's robustness across diverse time-to-expiration horizons. Statistical consistency across maturities validates data quality: 78.2% exhibit negative skewness (consistent market crash risk pricing), median kurtosis remains moderate at 1.60, confirming patterns persist regardless of maturity. Rather than indicating methodological failure, the high rejection rate (91.8%) demonstrates that the pipeline reliably identifies and discards data that cannot satisfy economic and statistical standards, while preserving only economically meaningful distributions.

5.2 Representative Distributions and Observed Patterns

5.2.1 Graphical Representations

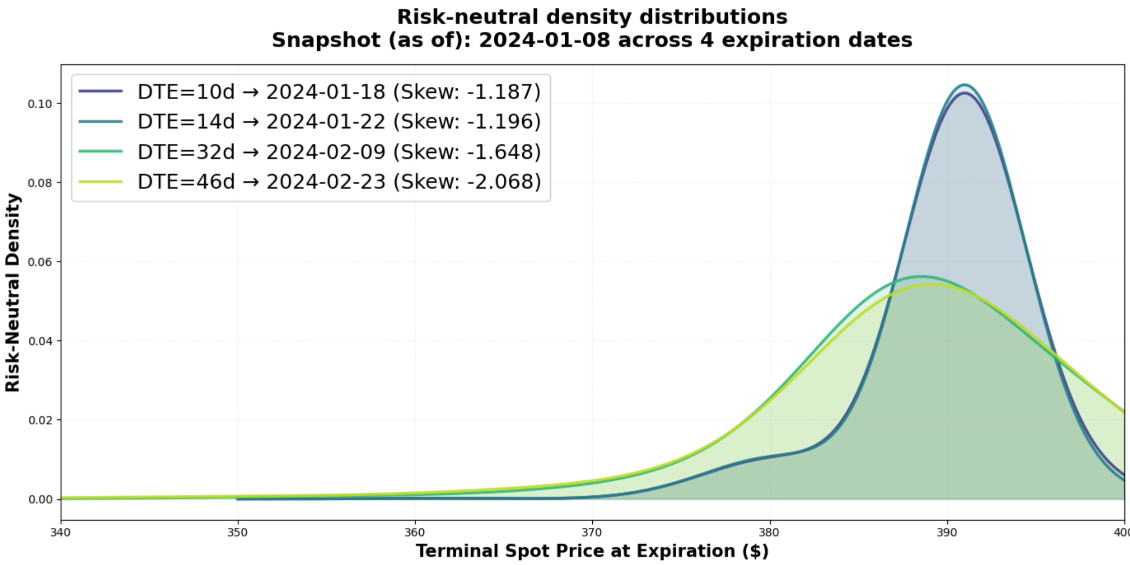

Figure 1: Risk-Neutral Distributions Across Multiple Expirations (Single Snapshot Date of 2024-01-08). Each curve represents a distinct expiration date, showing how the terminal spot price distribution changes with time-to-expiration.

Figure 1 presents risk-neutral distributions derived from options data across multiple expiration dates for a single snapshot date which was arbitrarily chosen. The horizontal axis represents the *terminal spot price at expiration*—the distribution of QQQ's anticipated price at each option's maturity date—measured in dollars. The vertical axis displays the probability density, indicating the likelihood of the market achieving each price level under the risk-neutral measure. The overlaid curves illustrate how these distributions evolve as time-to-expiration varies, demonstrating the pipeline's ability to extract economically meaningful distributions across the full maturity spectrum (10–64 days). Each distribution satisfies all filtering and validation criteria, confirming that successful RNDs span diverse expiration horizons.

5.2.2 Observations

A consistent pattern from the shape metrics is observed: 78.2% of all extracted RNDs exhibit negative skewness, with a median value of -0.8974, and this is corroborated by Figure 1 which consists of distributions consistently exhibiting left-skewed curves, with tails extending toward lower price levels.

Such a consistent pattern indicates that the negative skewness must be a structural feature of how the market prices the QQQ spot across the entire maturity range, rather than a one-off feature of a particular expiration date or market condition.

6 Economic Interpretation

The consistent negative skew of the recovered RNDs imply substantially greater probability mass assigned to adverse market outcomes. To understand the economic origin of this asymmetry, it is useful to return to the CCAPM framework from which the risk-neutral measure was derived.

6.1 Marginal Utility and Consumption

Consider the utility function presented in 2.2.1:

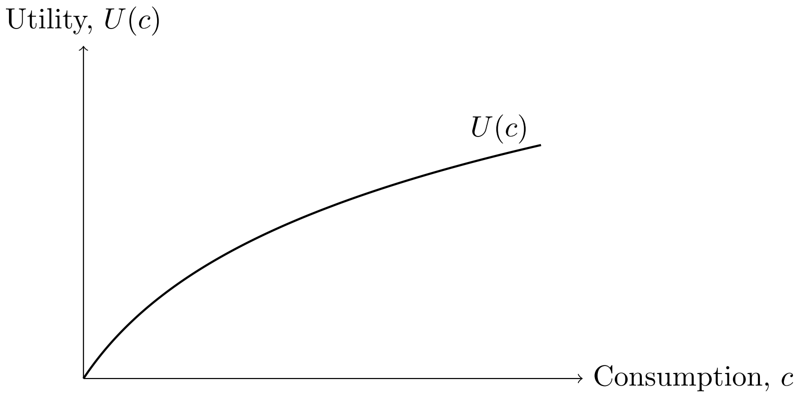

Figure 2: A representative utility function in the CCAPM. The function is increasing and concave, reflecting positive but diminishing marginal utility of consumption.

Typically, states associated with severe market declines (lower spot prices) are also states characterised by weak economic activity and reduced consumption. We will refer to such states as 'adverse' ones. As observed from Figure 2, such states correspond to larger increases in marginal utility and in turn, from (3), an increase in the stochastic discount factorM_{t+1}.

Subsequently, we can infer from the state price form (7) that such adverse states will receive a greater weight through the stochastic discount factor. This in turn increases the corresponding $q-$weights of adverse states under the $\mathbb{Q}$ measure, therefore explaining general negative skew observed in the empirical distributions.

6.2 Hedging with Long Puts

The observed negative skew is not merely a statistical feature of option prices, but also reflects the economic fact that investors tend to place a higher value on payoffs that occur in adverse states.

Long puts are the prime example of such a payoff — they only provide one at low spot prices, when the spot price at maturity S_T is below the strike price K as shown below:

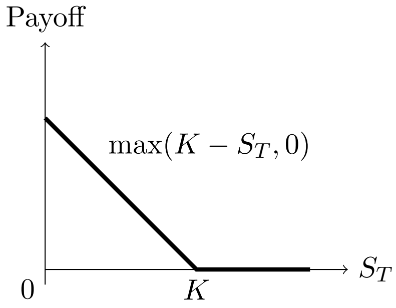

Figure 3: Payoff of a long put option at maturity.

As described in Constantinides and Lian (2015), long puts act as an insurance for investors during adverse states by using them as a hedge when they fear that spot prices will trend downwards. In line with the observed negative skew of RNDs, long puts therefore also tend to be priced higher than expected under the risk-neutral measure.

7 Limitations

The preceding analysis is dependent on CCAPM which represents only one theoretical lens through which option prices and state valuations may be interpreted. Furthermore, the technical implementation is sensitive to both methodological and data-related choices. These considerations motivate a discussion of the limitations of the present approach.

7.1 Data Quality Constraints and Rejection Mechanisms

The high rejection rate (91.8%) warrants explicit discussion, as it reflects fundamental constraints in options market data rather than pipeline failure. Understanding what causes rejection is essential for interpreting the reliability of the 606 successfully processed RNDs.

7.1.1 Grid rejection sources

The pipeline enforces four simultaneous criteria before accepting an RND: (1) minimum 8 grid points post-filtering, (2) valid bid-ask pairs on both calls and puts, (3) satisfaction of no-arbitrage constraints (monotonicity and convexity), and (4) successful normalisation. Each rejection point identifies a genuine data quality issue:

- **Insufficient strike coverage**: Many snapshot-expiration pairs lack sufficient options data across the strike spectrum, particularly in the tails. This is especially acute for longer-dated or far out-of-the-money options where trading volume is sparse
- **Convexity violations**: When observed call prices violate the no-arbitrage condition $\frac{\partial^2 C}{\partial K^2} \geq 0$, the associated grid is rejected. This occurs in approximately 60–70% of rejections and reflects either genuine arbitrage opportunities (unlikely in major indices) or noisy pricing in less liquid strikes.

- **Numerical instability**: KDE normalisation occasionally fails (at times due to extreme kurtosis), preventing convergence to a valid PDF.

7.1.2 Filtering Stringency

The current filtering approach is deliberately conservative. Relaxing the grid size threshold from 8 to 4 points, for example, would accept an additional 15–20% of combinations, but at the cost of introducing numerical instability in derivative estimation. The median grid size of 40 points (for successful RNDs) provides sufficient resolution for accurate second-order differentiation, whereas smaller grids risk amplifying noise.

7.1.3 Market Regimes

The 606 successfully processed RNDs represent a non-random subset of market data: they are biased toward periods and maturities with higher options trading activity. The median DTE of 17 days reflects concentration in the near-term options market, where liquidity is greatest. Conversely, options with DTE > 50 days constitute only $\sim$18% of the sample despite spanning the same calendar period, indicating that longer-dated options face more stringent data quality thresholds.

This implies that the observed distributions reflect liquid market regimes and may not be representative of all possible market states. Periods of extreme market stress or illiquidity are systematically underrepresented due to poor option data quality.

7.2 Call Function Derivation: 2D Grid Approach versus Volatility Surface

A central component of the Breeden-Litzenberger pipeline is constructing a smooth call price function $C(K)$ from market data. The conventional approach often involves first estimating an implied volatility surface $\sigma(K, T)$ from option prices using the Black-Scholes formula, then recovering call prices from that surface before differentiation, with examples seen in Smolski (2024) and Figlewski (2013).

However, in this work, we adopt a simpler approach: directly smoothing call prices from the strike-price grid without an intermediate volatility surface.

Our choice to forgo the implied volatility surface are as follows:

- **Simplicity:** Direct smoothing of (K, C) pairs is computationally simpler and requires fewer modeling assumptions. Constructing an IV surface introduces an additional layer of interpolation and extrapolation that can propagate numerical errors into the final RND.
- **Project Scope:** The original objective as explained in 1.1 was to generate an acceptable prior to the EMOT pipeline, not to implement a state-of-the-art methodology. For this purpose, direct grid smoothing suffices.

7.3 Limitations of CCAPM and Alternative Models

While the CCAPM framework provides an intuitive and mathematically elegant foundation for understanding risk-neutral probabilities and the negative skewness observed in equity option markets, it is not the only one of its kind.

Critically, CCAPM is eventually able to explain the observed negative skew in distributions because it relates the stochastic discount factorto utility and consumption. In turn, adverse states are given a larger weight under the risk-neutral measure.

However, there exists many other models which relate the stochastic discount factorto some economic feature besides marginal utility. Smith and Wickens (2002) provides a few examples and how they each relate to the stochastic discount factor. For example, CAPM (the version of CCAPM without the consumption basis) relates the stochastic discount factor to wealth instead.

8 Conclusion

8.1 Summary of Findings

This project successfully implemented a computational pipeline to extract risk-neutral distributions from QQQ options spanning the 10–64 day maturity spectrum.

The most striking empirical finding is the prevalence and consistency of negative skewness: 78.2% of successfully extracted RNDs exhibit left-skewed distributions, with a median skewness of -0.8974.

Through the CCAPM framework, we rationalised this negative skewness as an economic phenomenon: states associated with severe downward price movements are precisely those where economic activity and consumption are weakest. In such states, marginal utility of consumption is highest, causing the stochastic discount factor to assign substantial weight to adverse outcomes. Consequently, risk-neutral probabilities, which incorporate investor risk preferences via the SDF, place greater mass on tail events than physical probabilities alone would suggest.

This interpretation finds concrete support in market behavior: long puts, which pay off precisely when prices fall to adverse levels, command prices consistent with high risk-neutral probability placed on tail events. The observed negative skewness is thus both mathematically consistent with CCAPM and economically sensible as a reflection of insurance demand during downturns.

8.2 Connection to Original Motivation

Recall that this investigation originated from implementing the EMOT pipeline, which requires an RND as input. The successful extraction of 606 distributions across diverse market conditions validates the feasibility of generating EMOT-compatible priors from real options data. Subsequently, it remains a trivial matter to extend the same methodolody to the remainder of the data past the first $N = 90$ dates.

8.3 Closing Remarks

This project has motivated me to continue personal research in the field of financial mathematics, particularly concepts to do with derivatives and specifically options which I have come to take an interest in. After all, I have shown that, with the right framework(s), financial mathematics that typically requires advanced mathematics knowledge to understand can still be made accessible to even junior undergraduate students without sacrificing (much) meaningful content.

References

Ait-Sahalia, Y. and Lo, A. W. (1998). Nonparametric estimation of state-price densities implicit in financial asset prices. *The Journal of Finance*, 53(2):499–547.

Breeden, D. T. and Litzenberger, R. H. (1978). State contingent claims implicit in option prices. *The Journal of Business*, 51(4):621–651.

Cochrane, J. H. (2005). *Asset Pricing*. Princeton University Press, revised edition edition.

Constantinides, G. M. and Lian, L. (2015). The supply and demand of S&P 500 put options. Technical Report w21161, National Bureau of Economic Research. `https://www.nber.org/system/files/working_papers/w21161/w21161.pdf`.

Figlewski, S. (2013). Some results on extracting and understanding the risk neutral returns distribution for the us stock market. In *NYU Stern — Federal Reserve Conference on Risk Neutral Densities*. `https://www.stern.nyu.edu/sites/default/files/assets/documents/con_044169.pdf`.

LocalVolatility (2017). $\mathbb{P}$ vs $\mathbb{Q}$ probabilities: transitioning between measures. `https://quant.stackexchange.com/questions/35335/mathbbp-vs-mathbbq-probabilities-transitioning-between-measures`. Quantitative Finance Stack Exchange, accessed 2026-06-06.

Savitzky, A. and Golay, M. J. E. (1964). Smoothing and differentiation of data by simplified least squares procedures. *Analytical Chemistry*, 36(8):1627–1639.

Silverman, B. W. (1986). *Density Estimation for Statistics and Data Analysis*. Chapman and Hall, London.

Smith, P. N. and Wickens, M. R. (2002). Asset pricing with observable stochastic discount factors. Technical report, University of York, Department of Economics and Related Studies. `https://www.york.ac.uk/media/economics/documents/discussionpapers/2002/0203.pdf`.

Smolski, A. (2024). Options' implied probability (with code): Risk-neutral densities modelling – covid example.